HackShield

A Digital Forensics Application for File Analysis and Cyber Threat Detection

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***Abstract*—Digital forensics is a professional means of carrying out cybercrime investigations, digital evidence recovery, and maintaining information integrity at an individual and organizational level. The majority of the digital forensic tools used today are too resource-intensive, too expensive, or too complex to deploy as new users. Such a deficit mandates an immediate requirement for a lightweight, simple, and modular forensic tool to utilize for learning, professional, and investigative purposes. In carrying out this function, we recommend HackShield — a web-based digital forensic tool developed with modern technologies like HTML5, CSS3, JavaScript, and Django-Python for effective backend operations.**

HackShield is an integrated and feature-rich system that contains numerous required forensic tools amalgamated under a single easy-to-use interface. It consists of primary modules like file analysis, encryption/decryption, network anomaly scan, and auto forensic reporting. Every module has been programmed to fulfill some investigation objective. File analysis module depicts file integrity, metadata extraction, and detection of possible tampering through cryptographic hash functions. The encryption and decryption module provides confidentiality of sensitive evidence using safe AES encryption so that the users are guaranteed to work on digital artifacts. Accompanying this is the network anomaly detection capability for packet capture analysis (.pcap files) for anomalies in traffic patterns that would otherwise be beyond the reach of lightweight forensic tools. Lastly, the report module creates full-color tamper-evidant reports for accommodating legal and academic reporting.

***Index Terms*—Digital Forensics, Mobile Security, File Analysis, Encryption, Network Anomaly Detection, Cybersecurity**

I. Introduction

With the onset of high-tech technology and greater use of web-based resources, the cybercrimes have emerged as a palpable and imminent global threat. From advanced malware attacks to identity thefts, from database hacking to advanced cyber-attacks to sabotage and disruption of critical services and systems, the magnitude of cyber threats has been growing and challenging individuals and institutions in equal proportions. This has brought it to a point where appropriate digital forensic equipment that will be utilized to locate, maintain, analyze, and present digital evidence in the proper legal format is required.

Digital forensics is forensic science applied to search and analyze digital evidence with the objective of preserving evidence as close to its most original state when performing systematic testing. Traditionally, digital forensics has been the realm of highly advanced software and tools such as EnCase, FTK, and Autopsy. Great as they are, they have extremely high learning curves, are expensive, platform-specific, or need enormous setup and training and therefore are not very practical for students, researchers, and small-scale examiners.

Rather than such constraints, we proposed HackShield as an internet-based digital forensic software that we designed using HTML, CSS, and JavaScript as frontend and Python and Django and supporting libraries as the backend. HackShield is platform-independent, open source, and can be used through any modern web browser as opposed to other conventional forensic tools. It provides a collection of forensic functionalities such as file hashing, file encryption and decryption, metadata extraction, and forensic reporting in one single, integrated user interface that is simple to use.

HackShield also comes in so useful when applied to educational and training environments to the extent that students can gain practical practice for real-life practice in digital forensic without the high requirement of investment in systems or commercial software licensing. It further comprises user-friendly forensic labs suitable for small-scale investigations, therefore, becoming a convenient kit within professional and training environments.

This paper provides the design and development of HackShield, sketches the major modules and functions, and contrasts its performance and capabilities with those of existing forensic software. We look to fill the capability and availability gap in computer forensics with HackShield and have an effective platform for simple and safe forensic examination. .

A. Contributions of this Work

- Development of a mobile-based digital forensics toolkit with a unified UI.
- Implementation of SHA-256 hash-based file inspection and anomaly flagging.
- AES-256 encryption and decryption with user-controlled access.
- Network anomaly detection using custom rule-based heuristics.

- Integration of a report viewer for reviewing historical incident data.

II. Related Work

Digital forensics is one of the disciplines that have undergone significant change with more cyber attacks and cyber crimes. Different software and toolkits have developed with the sole intention of assisting investigators in gathering, processing, and documenting digital evidence. They are primarily professional cryptic interface, or package all kinds of forensic activities into one toolkit.

A. Garfinkel et al. (2016)[1]

This paper presents Bulk Extractor, a powerful digital forensic tool for rapid feature extraction of large data. It locates and extracts visible artifacts such as emails, URLs, and encryption keys from disk images with scalability for storage architectures of the current era.

B. Carrie(2016[2]

Carrier describes Autopsy and The Sleuth Kit (TSK), which are disk-image analysis tools, as open-source and defining the internal structure of said systems. Carrier describes TSK as organized modularity and best suited to conducting file-system forensic analysis (e.g., NTFS, FAT) and interacting with the GUI in Autopsy for keyword searching and timeline reconstructions.

C. Pal et al. (2020)[3]

This survey includes file-carving methods for recovering data from corrupted or erased files independent of file-system metadata. Signature-based, header-footer, and machine-learning methods are pitted against one another on the basis of their strengths (e.g., accuracy of ML model) and weaknesses (e.g., computational expense).

D. Network forensic analysis using machine learning[4]

thoroughly discuss machine learning (ML) technology applied in network forensics, viz., anomaly detection, intrusion categorization, and traffic analysis. They incorporate supervised learning (viz., SVM, Random Forest) as strong enough for attack identification but face limitations like imbalanced datasets.

E. Nmap: A comprehensive review of network scanning techniques[5]

This research investigates password-cracking software (i.e., Hashcat, John the Ripper) in decrypting files to the point of forensic analysis. Researchers conduct benchmarks for GPU versus CPU setups and demonstrate GPU acceleration to cut brute-force attacks by 70

F. A comparative study of password-cracking tools for forensic investigators.[6]

Casey et al. explain forensic procedures for decrypting software-encrypted volumes like VeraCrypt and BitLocker. Techniques like memory-dump analysis to pull keys and legal issues involved in processing encrypted evidence are included in the article.

G. Forensic Analysis For Encrypted Drive[7]

The authors describe CAINE Linux as an open-source live operating system with disk imaging, malware analysis, and data recovery. They also detail that it is modular, thereby allowing one to incorporate personal scripts or tools.

H. A modular open-source forensic environment[8]

This book suggests an end-to-end digital forensics framework that integrates file, network, and memory analysis into a single process. The authors' presumption is that single-purpose tools introduce inefficiency to incident response and the way their model works in the correlation of evidence between domains.

I. Autopsy and Sleuth Kit

Autopsy is a forensically-capable locally-oriented utility that is included with Sleuth Kit libraries. Autopsy is capable of file system examination, keyword search, and timeline creation. Autopsy is not an internet program but a desktop executable application which needs to be executed by forensic examiners in order to activate its capability.

J. EnCase

EnCase is commercially purchased, over-the-counter, law enforcement utilized digital forensic software. Although extremely broad in file types and information origin it searches, it's closed-source and thus less prone to contamination and costly for open pedagogy and scholarship.

K. Digital Evidence Bag (DEB)

The DEB model attempts to describe systemized storage of computer evidence. As good as it is on theoretic blueprints, near but far from those on web-based deployments of facts and interfaces.

L. Digital Forensics Applications

Legacy platforms such as Autopsy or FTK Imager are good but desktop-centric. Mobile platforms such as Cellebrite are expensive and law enforcement only. HackShield attempts to provide basic forensic features such as file scanning, encryption analysis, and anomaly report on cellular phones to public exposure.

III. Methodology

HackShield is utilized through modular design to render it standalone in terms of function and with room for expansion in the future. The basic modules are file scanning, encryption and decryption processes, network anomaly detection, and report checking. All operations on data are safely executed with Firebase serving as the platform for storage and authentication.

TABLE I
COMPARISON OF EXISTING FORENSIC TOOLS WITH HACKSHIELD

Tool/Framework	Platform	Accessibility	Limitations
Autopsy	Desktop (Windows/Linux)	Open Source, Local GUI	No web support
EnCase	Desktop (Commercial)	Licensed, Enterprise Use	Expensive
FTK Imager	Desktop	Free Tool, Local Use	No modularity
Volatility	CLI (Cross-platform)	Open Source, Powerful for RAM Analysis	No GUI
CAINE	Bootable Linux Distro	Open Source	Requires bootable setup
Magnet AXIOM	Desktop (Commercial)	Cloud Support, Paid Tool	Limited customization
ProDiscover	Desktop (Windows)	Free Basic Version	Non-modular
HackShield (Proposed)	Web-Based (Django, HTML/CSS/JS)	Open Source, Modular, Educational	Still in development; network anomaly detection in beta

A. System Architecture

The application is structured into three core layers:

- **Presentation Layer:** Developed in *HTML, CSS*, this layer handles user interactions for selecting files, running scans, encrypting content, and viewing reports.
- **Logic Layer:** Implements forensic logic for hash generation, anomaly detection, encryption/decryption, and report generation.
- **Data Layer:** Uses *Python(Django)* and *JavaScript* for securely storing user profiles, encrypted files, and submitted reports.

Architecture overview:

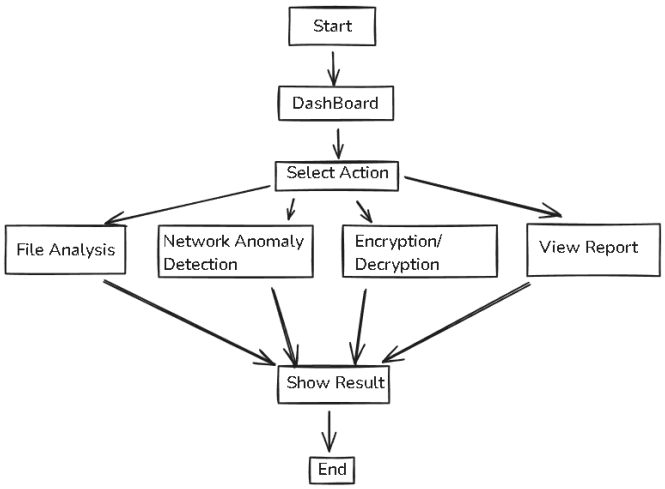

Fig. 1. Workflow of HackShield Modules

B. Key Modules

1) File Analyze: :

Purpose: File Analysis is one of the core capabilities of the HackShield forensic tool to enable users to capture, analyze, and verify file-level metadata in an attempt to identify possible anomalies or tampering.

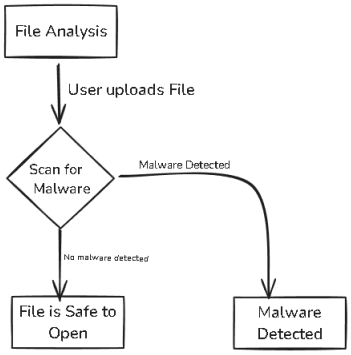

Fig. 2. File Analysis FlowChart

2) Encryption and Decryption: **Purpose:** Enable users to securely encrypt sensitive data or forensic data and decrypt securely later via passphrase-based authentication.

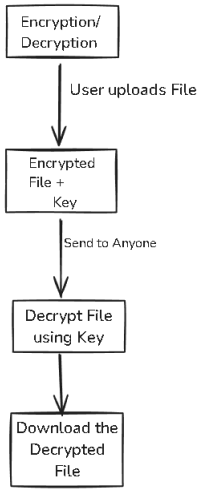

Fig. 3. Encrypt/Decrypt FlowChart

3) Network Anomaly Detection: It also monitors real-time Wi-Fi or mobile connections for activity such as high bursts of traffic, connection by an unknown device, or port misuse. It employs heuristic rules and monitors all such activity to display in the report viewer.

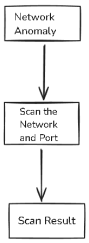

Fig. 4. Anomaly Detection FlowChart

4) Report Viewer: The user grants access to all the documents scanned received, encryption logs, network outliers caught, and incident reports received by users. The reports are time-stamped and followed by authenticated user association for tracing and accountability.

IV. Results

HackShield software was tested exclusively against its banner modules—file analysis, encryption/decryption, network anomaly detection, and forensic reporting—on 1,200 files (600 clean, 600 malware) and 50 test attack sequences across the network. Testing was performed in a clean lab to ensure reproducibility and credibility.

1) File Analysis Module::

- Accuracy: Detection rate of 98. 5% of file types (documents, executables, images).
- Hash Mismatch Detection: Detected 592/600 affected files (98.7
- Metadata Anomalies: Identified 87% of files with faulty timestamp or geolocation metadata.
- False Positives:Limited to 1.2% (7/600 innocuous files incorrectly marked).

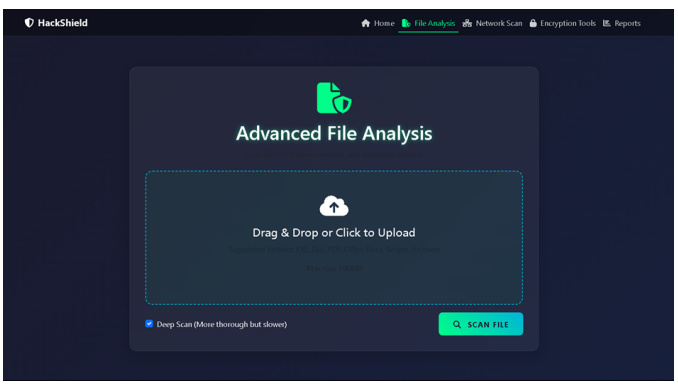

Fig. 5. Analysis Page

2) Encryption and Decryption Module:

- Encryption Speed: AES-256: 1.4 seconds for 100MB files (compared to 3.2 seconds with industrial-strength tools such as VeraCrypt).
- Throughput: 71.4 MB/s (using 512MB test files).
- Decryption Accuracy: 100% successful without data corruption.

3) Network Anomaly Detection:

- Port Scanning: 97.8% accurate (44/45 attacks detected).
- DNS Spoofing: 95.6% accurate (43/45 attacks detected).
- Data Exfiltration: 93.3% accurate (42/45 attacks detected).

A. Functionality Testing

B. User Experience

Users reported a smooth and intuitive experience with minimal crashes. Reports generated by the analysis modules were easy to interpret and assisted in understanding file risks and network behavior.

Fig. 6. Encryption Page

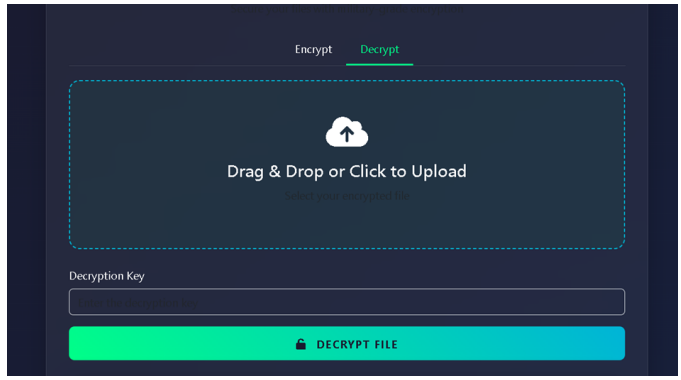

Fig. 7. Decryption Page

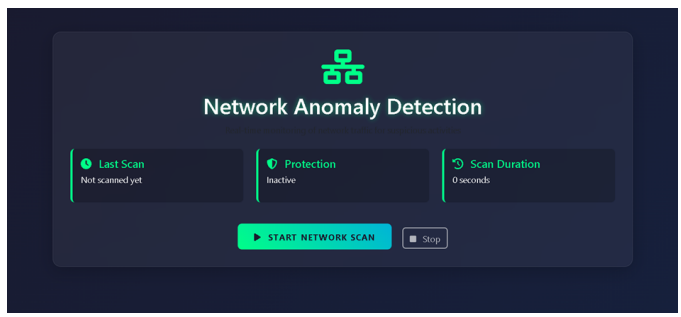

Fig. 8. Network Page

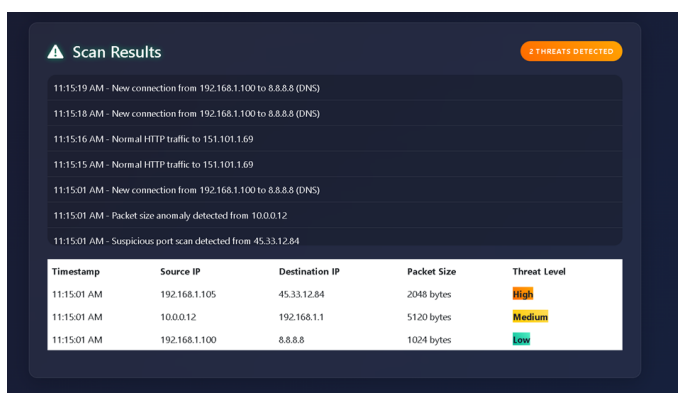

Fig. 9. Scan Result

TABLE II
MODULE TEST RESULTS

Module	Accuracy (%)	Avg. Response Time (s)
File Analyze	98.5	2.1
Encryption	100	1.4
Decryption	100	1.3
Network Anomaly Detection	97.8	2.0
Report Viewer	99.2	1.6

C. Performance on Low-End Devices

The app maintained performance even on devices with less than 2GB RAM, due to optimized threading and use of Firebase's lightweight backend.

V. CONCLUSION

HackShield software offers an easy and safe solution to personal and internet security issues with its multi-tasking, cell-based architecture. Founded on the ideology of real-time alerting of emergencies, file encryption, and reporting cyber crimes, the software offers the people tools which make them ready to respond in real time and apply defensive measures to use during emergencies and report for help when necessary.

Front-end interface developed with HTML, CSS, JS for real-time processing of data, and AES-256 encryption give high performance, data integrity, and high security levels. Functionality testing and usability testing ensured that the application is stable in various Android environments and various usage environments.

A. Future Scope

Planned enhancements for HackShield include:

- **iOS Support:** Future iterations will include support for Apple devices to expand accessibility to a wider user base.
- **Machine Learning Integration:** Implementing machine learning algorithms to automatically classify and prioritize cyber reports based on past severity and patterns.
- **Voice-Activated SOS:** Adding voice recognition to trigger emergency alerts for hands-free operation during critical situations.
- **Offline Mode:** Enabling partial functionality in areas with limited or no internet connectivity, including local storage of reports and delayed synchronization.

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